

16.07: Dynamics

Fall 2026

Course Description

This course is about modeling and simulating mechanical systems, starting with a review of basic Newtonian mechanics of point-mass particles, then covering rigid-body and multi-body systems. Applications to aircraft, spacecraft, and robotic systems will be emphasized.

Prerequisites: Officially – (16.001 or 16.002) and (16.003 or 16.004). Practically – basic mechanics at the level of 8.01, calculus and vectors/matrices at the level of 18.02, and prior experience with a high-level programming language like Python or MATLAB (we'll use MATLAB).

Instructors

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TA: John Zhang	Email: jzhang3@mit.edu	

Logistics

- Lectures will be held **Tuesdays and Thursdays 11:00–12:30 in 35-225**. Lectures will also be recorded for later viewing.
- Recitations and labs will be held **Fridays 3:00–4:00 in **TODO: TBD****.
- Office hours will be **TODO: TBD**.
- Homework assignments will be due **Fridays by 5:00 PM**.
- There will be occasional short (5 minute) quizzes given in class.
- There will be an in-class mid-term exam and a cumulative final exam.
- Piazza will be used for general discussion and Q&A outside of class and office hours.

Learning Objectives

By the end of this course, students should be able to do the following:

1. Select appropriate coordinates to describe the motion of particle and multi-body system
2. Formulate dynamics in non-inertial reference frames
3. Identify and exploit conserved quantities like energy and momentum
4. Derive equations of motion for multi-body systems using the Euler-Lagrange equation
5. Model the orbital dynamics of a spacecraft as central-force motion of a particle
6. Use Euler's equation to model and analyze the attitude dynamics of a rigid body and apply this to aircraft and spacecraft

Learning Resources

There is no required textbook for this course. Video recordings of lectures and lecture notes will be posted online. The following books are good references for the topics discussed in the course that you may want to refer to:

1. J. L. Meriam, L. G. Kraige, and J. N. Bolton, Engineering Mechanics: Dynamics, 9th ed. Hoboken, NJ, USA: Wiley, 2020.
2. L. D. Landau and E. M. Lifshitz, Mechanics, 3rd ed. Oxford, U.K.: Butterworth-Heinemann, 1976.
3. H. Goldstein, C. Poole, and J. Safko, Classical Mechanics, 3rd ed. San Francisco, CA, USA: Addison Wesley, 2002.
4. K. M. Lynch and F. C. Park, Modern Robotics: Mechanics, Planning, and Control. Cambridge, U.K.: Cambridge University Press, 2017.

Assignments

- Students will be asked to complete homework assignments roughly weekly that include both an analytical question and a numerical simulation question to be completed in MATLAB.
- There will be three labs throughout the semester during the Friday recitation time. On lab weeks, homework assignments will include analysis of experiment data.

Quizzes and Exams

- Short quizzes will be given approximately weekly in class. These are intended to take no more than 5 minutes to complete.
- There will be an in-class mid-term exam and a cumulative final exam.

Grading

Grading will be based on:

- 25% Homework and Labs
- 15% Quizzes
- 25% Mid-Term Exam
- 30% Final Exam
- 5% Participation

Students can earn a full participation grade through any combination of office hours, Piazza discussions, recitation attendance, and by offering constructive feedback about the course to the instructors.

Course Policies

Late Assignments: Students are allowed a budget of six late days for turning in homework with no penalty throughout the semester. They may be used together on one assignment, or separately across multiple assignments. Beyond these six days, no other late homework will be accepted.

Collaboration: Students are encouraged to collaborate on labs and homework assignments, but each student must turn in their own assignment by the deadline.

AI Use: Students are permitted to use AI tools like Claude and Gemini for assistance on homework assignments. This should be acknowledged explicitly with a brief statement about how AI was used.

Tentative Schedule

Week	Dates	Topics	Assignments
1	Sep 10	Course Overview, History, and Review	Survey
2	Sep 15 Sep 17	Coordinates and Reference Frames Momentum, Energy, and Conservation	HW1 Out
3	Sep 22 Sep 24	Euler-Lagrange Equations Potential Functions, Non-Conservative Forces	
4	Sep 29 Oct 1	Simulation and Runge-Kutta (Zac Out) Constraints and Lagrange Multipliers	HW1 Due HW2 Out
5	Oct 6 Oct 8	Orbit Mechanics 1 Orbit Mechanics 2	
6	Oct 13 Oct 15	No Class — Monday Schedule Rigid Bodies and Attitude	HW2 Due HW3 Out
7	Oct 20 Oct 22	Rotation Matrices and Quaternions Euler's Equation	
8	Oct 27 Oct 29	Newton-Euler Dynamics for Aircraft Mid-Term Exam (Zac Out)	
9	Nov 3 Nov 5	Linearization and Stability (Zac Out) Linear Dynamical Systems and Modes (Zac Out)	
10	Nov 10 Nov 12	Vector Fields and Phase Portraits Coulomb Friction and the Maximum Dissipation Principle	HW3 Due HW4 Out
11	Nov 17 Nov 19	3D Multi-Body Dynamics Simulation Constraint Stabilization	
12	Nov 24 Nov 26	System Identification and Least-Squares No Class — Thanksgiving	HW4 Due HW5 Out
13	Dec 1 Dec 3	TBD: Schedule Margin	
14	Dec 8 Dec 10	TBD: Schedule Margin	HW5 Due